

The clock component will repeat a particular action after a fixed duration of time, cyclically.

7. **Horizontal arrangement:** This keeps all the child components horizontally aligned.
8. **Notifier:** This is used to display some instructions or give controls over existing components. You will see its functionality in more detail as we will be implementing it in our game.

The table on the right shows the components that we will require for this application. We will drag them on to the designer from the left hand side palette.

1. Drag and drop the components mentioned in the table on the right to the viewer.
2. Visible components can be seen, while the non-visible components will be located beneath the viewer under the tag 'Non-visible'.
3. We have used labels to identify applications.
4. All buttons need to be put within the *Horizontal arrangement* to keep them aligned horizontally.
5. If you have dragged and placed everything, the layout will look something like what's shown in Figure 1.
6. Make the necessary property changes like we did while changing the text property for the Label and Button components.
7. Renaming the components helps to identify them in the block editor.
8. Your graphical user interface is now ready. Figure 1 shows how the application will look after the installation.
9. Figure 2 shows the hierarchy of the components that we have dragged to the designer.

If you are confused after seeing the designer and the components viewer, let me explain it a bit more. Here is the hierarchy that we have placed for our application.

1. At the top, we have a horizontal arrangement, which has four label components as its child components. Of these, two will be

	Component name	Purpose	Location
1	Canvas	To let the graphical components be visible	Palette-->Drawing and Animation-->Ball
2	Label	To display a label	Palette-->User Interface-->Label
3	Ball	Displays the ball on the canvas	Palette-->Drawing and Animation-->Ball
4	Clock	To check the status at fixed intervals	Palette-->Sensors-->Clock
5	Button	To trigger events	Palette-->User Interface-->Button
6	Horizontal arrangement	To arrange the child components	Palette-->Layout-->Horizontal Arrangement
7	Image sprite	To display graphical images	Palette-->Drawing and Animation-->Image Sprite
8	Notifier	To display on-screen information	Palette-->User Interface-->Notifier

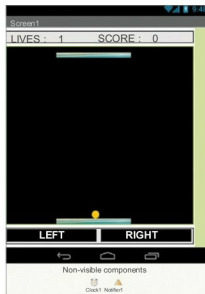


Figure 1: Designer screen

displayed as headings while the other two will be dynamic in nature, which means that their value will change according to the behaviour. For each of the label components, the font is set at 20 points and width is set to fill the parent.

2. Next, we have the canvas component with the width and height to fill the parent. Within this, we have a ball component and two image sprites, namely 'Upper_paddle_sprite' and 'Lower_paddle_sprite'.
3. For the ball, we have set the properties given in Figure 3 in the

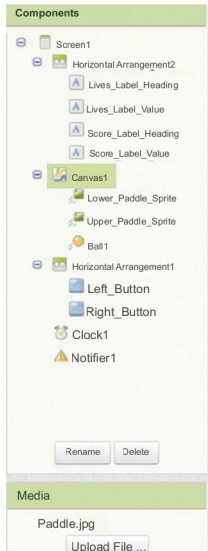


Figure 2: Components view